



Yakeen Plus (2025)

Practice Test - 13

DURATION : 200 Minutes

DATE : 09/12/2024

M. MARKS : 720

Topics Covered

Physics :	Optics Dual Nature of Matter and Radiation Atoms and Nuclei Electronic Devices
Chemistry :	Chemical Equilibrium, Ionic Equilibrium, Structure of Atom, Amines, Biomolecules, Practical Organic Chemistry, Salt Analysis
Biology :	(Botany) : Molecular Basis of Inheritance, Organisms and Population, Ecosystem, Biodiversity and Conservation (Zoology) : Human Health and Diseases; Biotechnology: Principles & Processes; Biotechnology and its Applications

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours 20 min.** duration.
3. The test booklet consists of **200** questions. The maximum marks are **720**.
4. There are four Section in the Question Paper, Section I, II, III & IV consisting of Section-I (**Physics**), Section-II (**Chemistry**), Section-III (**Botany**) & Section IV (**Zoology**) and having **50 Questions** in each Subject and each subject is divided in two Section, **Section A** consisting 35 questions (all questions are compulsory) and **Section B** consisting 15 Questions (Any 10 questions are compulsory).
5. There is only one correct response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ Invigilator's Signature _____

SECTION-(I) PHYSICS

SECTION - A

- Consider 3^{rd} orbit of He^+ (Helium) using non relativistic approach the speed of electron in this orbit nearly is:
 (1) $1.46 \times 10^6 m/s$ (2) $0.73 \times 10^6 m/s$
 (3) $3.0 \times 10^8 m/s$ (4) $2.92 \times 10^6 m/s$
- An illuminated point object is placed on the principal axis, in front of an equi-convex glass lens of focal length $f = 40$ cm, at a distance of $u = 1.20$ m from the lens. A reflecting plane mirror has been placed behind the lens perpendicular to the principal axis and facing the lens. The nature of the final image and the distance of the plane mirror from the lens, so as to form the final image at the plane mirror itself, is
 (1) Virtual image, 40 cm
 (2) Real image, 40 cm
 (3) Virtual image, 50 cm
 (4) Real image, 20 cm
- If the difference between $(n+1)^{th}$ Bohr radius and n^{th} Bohr radius is equal to the $(n-1)^{th}$ Bohr radius then find the value of n
 (1) 4 (2) 3
 (3) 2 (4) 1
- If an α - particle of mass m , charge q and velocity v is incident on a nucleus of charge Q and mass m , then the distance of closest approach is
 (1) $\frac{Qq}{4\pi\epsilon_0 m^2}$ (2) $\frac{Qq}{2\pi\epsilon_0 m v^2}$
 (3) $\frac{Qq m v^2}{2}$ (4) $\frac{Qq}{m v}$
- In photoelectric effect if the intensity of light is doubled, then maximum kinetic energy of photoelectrons will be:
 (1) double (2) half
 (3) four times (4) no change
- An electron changes its position from orbit $n = 4$ to the orbit $n = 2$ of an atom. The wavelength of the emitted radiations is ($R =$ Rydberg's constant)
 (1) $\frac{16}{R}$ (2) $\frac{16}{3R}$
 (3) $\frac{16}{5R}$ (4) $\frac{16}{7R}$

- The de Broglie wavelength of an electron in the n^{th} Bohr orbit is related to the radius R of the orbit as

$$\begin{aligned} (1) \quad n\lambda &= \pi R & (2) \quad n\lambda &= \frac{3}{2}\pi R \\ (3) \quad n\lambda &= 2\pi R & (4) \quad n\lambda &= 4\pi R \end{aligned}$$

- r_1 and r_2 are the radii of atomic nuclei of mass numbers 64 and 27 respectively. The ratio (r_1 / r_2) is

$$\begin{aligned} (1) \quad 64 / 27 & & (2) \quad 27 / 64 \\ (3) \quad 4 / 3 & & (4) \quad 1 \end{aligned}$$

- Match the following lists:

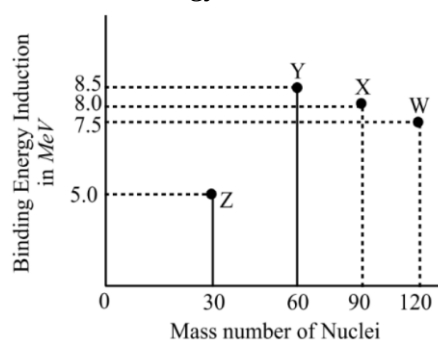
List-I		List-II	
P.	Photoelectric effect	1.	Photon
Q.	Wave	2.	Frequency
R.	X-rays	3.	K capture
S.	Nucleus	4.	g-rays
		5.	None of these

- P-1, Q-2, R-3, S-4
- P-2, Q-1, R-3, S-4
- P-1, Q-2, R-4, S-3
- P-4, Q-2, R-3, S-1

- A nucleus ${}_Z X^A$ emits an α -particle with velocity v . The recoil speed of the daughter nucleus is

$$\begin{aligned} (1) \quad \frac{A-4}{4v} & & (2) \quad \frac{4v}{A-4} \\ (3) \quad v & & (4) \quad \frac{v}{4} \end{aligned}$$

- Binding energy per nucleon vs mass number curve for nuclei is shown in figure. W, X, Y and Z are four nuclei indicated on the curve. The process that would release energy is



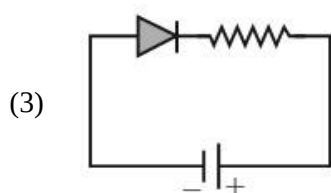
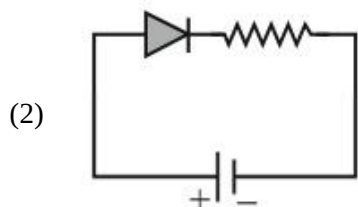
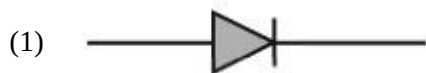
- Y $\rightarrow$ 2Z (2) W $\rightarrow$ X + Z
 (3) W $\rightarrow$ 2Y (4) X $\rightarrow$ Y + Z

12. One milligram of matter converted into energy, will give
 (1) 9 joule (2) 9×10^3 joule
 (3) 9×10^5 joule (4) 9×10^{10} joule
13. When two nuclei of mass X and Y respectively fuse to form a nucleus of mass m along with the liberation of some energy, then-
 (1) $X + Y > m$ (2) $X - Y = m$
 (3) $X + Y = m$ (4) $X + Y < m$
14. The maximum kinetic energy of photoelectrons emitted from a surface when photons of energy 6eV fall on it is 4eV. The stopping potential in volt is
 (1) 2 (2) 4
 (3) 6 (4) 10
15. If the kinetic energy of the particle is increased by 16 times, the percentage change in the de Broglie wavelength of the particle is:-
 (1) 25% (2) 75%
 (3) 60% (4) 50%
16. In a positron decay, the radionuclide ^{11}C decays according to $^{11}\text{C} \longrightarrow ^{11}\text{B} + e^+ + \nu$ [the respective atomic masses are $^{11}\text{C} = 11.01142 \text{ u}$, $^{11}\text{B} = 11.00931 \text{ u}$ and the mass of positron = the mass of electron = $m_e = 0.00055 \text{ u}$]. The disintegration energy (i.e. the Q value) is approximately
 (1) 1.45 MeV (2) 0.94 MeV
 (3) 0.43 MeV (4) -1.45 MeV
17. In an n-type silicon, which of the following statement is true?
 (1) Electrons are majority carriers and trivalent atoms are the dopants.
 (2) Electrons are minority carriers and pentavalent atoms are the dopants.
 (3) Holes are minority carriers and pentavalent atoms are the dopants.
 (4) Holes are majority carriers and trivalent atoms are the dopants.
18. At a certain location on the Earth surface, the intensity of sunlight is 1.00 kW/m^2 . A perfectly reflecting concave mirror, of radius of curvature R and aperture radius r, is facing the Sun to produce the light intensity of 100 kW/m^2 at the image. Knowing that the disc of the Sun subtends an angle of 0.01 radian at the Earth surface, the relation between R and r is
 (1) $R = 10r$ (2) $R = 20r$
 (3) $R = 40r$ (4) $r = 20R$
19. Which statement is **correct**:
 (1) N-type germanium is negatively charged and P-type germanium is positively charged
 (2) Both N-type and P-type germanium are neutral
 (3) N-type germanium is positively charged and P-type germanium is negatively charged
 (4) Both N-type and P-type germanium are negatively charged
20. When the two-plane mirror are parallel to each other, then the number of images of an object placed between them is
 (1) 2 (2) 3
 (3) 4 (4) Infinite
21. The energy band gap is maximum in:
 (1) Metals (2) Superconductors
 (3) Insulators (4) Semiconductors
22. An illuminated point object is placed in front of an equi-convex lens of refractive index $\mu = 1.5$ and focal length $f = 40 \text{ cm}$, at a distance of $u = 1.20 \text{ m}$ in front of the lens on its principal axis. Water ($\mu = 4/3$) fills the space behind the lens up to a distance of 40 cm from the lens. The final image is formed on the principal axis beyond the lens at a distance v from the lens. The value of v and the nature of image are
 (1) 60 cm, virtual (2) 80 cm, virtual
 (3) 110 cm, real (4) 130 cm, real
23. **Assertion A:** The focal length of the convex mirror will increase, if the mirror is placed in water.
Reason R: The focal length f of a convex mirror of radius R is equal to $R/2$.
 (1) Both **Assertion (A)** and **Reason (R)** are the true, and **Reason (R)** is a correct explanation of **Assertion (A)**.
 (2) Both **Assertion (A)** and **Reason (R)** are the true, but **Reason (R)** is not a correct explanation of **Assertion (A)**.
 (3) **Assertion (A)** is true, and **Reason (R)** is false.
 (4) **Assertion (A)** is false, and **Reason (R)** is true.

24. According to Bohr theory, in the ground state of hydrogen atom, an electron revolves in circular orbit of radius r with velocity v and circulation frequency f . The magnetic dipole moment of the electronic orbit is μ_m . The magnetic field produced by the circulating electron at the center of the atom is B . Then for a He^+ ion in the state $n = 2$ (electron in the 2nd orbit)

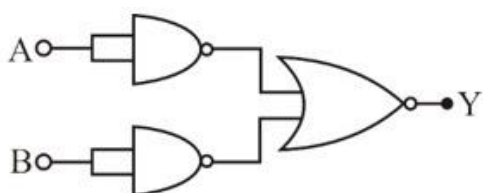
- (1) the radius of the orbit is $\frac{r}{2}$
- (2) the magnetic dipole moment of the orbit is $4\mu_m$
- (3) the frequency of circulation is $\frac{f}{4}$
- (4) the magnetic field at the center is $\frac{B}{4}$

25. Which one is in forward bias



- (4) None of these

26. The operation performed by the circuit as shown in figure, is

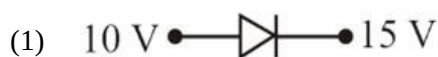


- (1) OR
- (2) NOR
- (3) AND
- (4) NAND

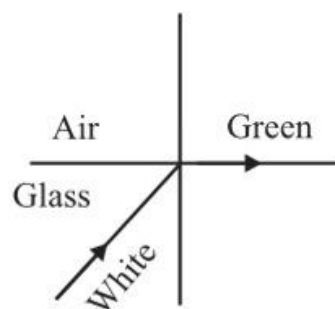
27. Which of the following is **not** due to total internal reflection?

- (1) Working of optical fibre
- (2) Difference between apparent and real depth of a pond
- (3) Mirage on hot summer days
- (4) Brilliance of diamond

28. For the diodes given in figures below, the forward biased diode is

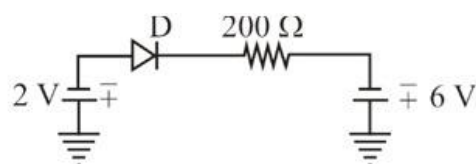


29. White light is incident on the interface of glass and air as shown in figure. If green light is just totally internally reflected then the emerging ray in air contains:



- (1) Yellow, Orange, Red
- (2) Indigo, Blue, Violet
- (3) All colours
- (4) All colours except green

30. Find the current flowing through diode D

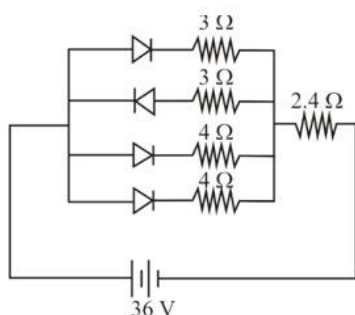


- (1) 20 mA
- (2) 10 mA
- (3) Zero
- (4) 40 mA

31. A thin glass (refractive index 1.5) lens has optical power of -5 D in air. Its optical power in liquid with refractive index 1.6 will be
- (1) -1 D (2) $+1\text{ D}$
 (3) -25 D (4) $+25\text{ D}$

32. What should be the angle between two plane mirrors, if an object placed unsymmetrically between them has five images?
- (1) 72° (2) 50°
 (3) 120° (4) 59°

33. In the given circuit, voltage across $2.4\ \Omega$ resistor is



- (1) Zero (2) 12 V
 (3) 36 V (4) 24 V
34. If an observer is walking away from a stationary plane mirror with 6 m/sec . Then the velocity of the image with respect to observer will be
- (1) 6 m/sec (2) -6 m/sec
 (3) 12 m/sec (4) 3 m/sec
35. An object is placed at a distance of $f/2$ from a convex lens. The image will be
- (1) At one of the foci, virtual and double its size
 (2) At $3f/2$, real and inverted
 (3) At $2f$, virtual and erect
 (4) None of these

SECTION - B

36. In LED and photodiode, the p-n junction is (respectively)
- (1) Reverse bias and forward bias
 (2) Forward bias and reverse bias
 (3) Reverse bias and reverse bias
 (4) Forward bias and forward bias

37. Which of the following breakdown of P-N junction is reversible?
- (1) Avalanche breakdown
 (2) Zener breakdown
 (3) Dielectric breakdown
 (4) All of these

38. An electron (mass m) with initial velocity vector $\mathbf{v} = v_0\mathbf{i} + v_0\mathbf{j}$ is in an electric field vector $\mathbf{E} = -E_0\mathbf{k}$. If λ_0 is initial de-Broglie wavelength of electron, its de-Broglie wave length at time t is given by:

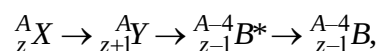
(1) $\frac{\lambda_0\sqrt{2}}{\sqrt{1 + \frac{e^2 E_0^2 t^2}{m^2 v_0^2}}}$ (2) $\frac{\lambda_0}{\sqrt{2 + \frac{e^2 E_0^2 t^2}{m^2 v_0^2}}}$
 (3) $\frac{\lambda_0}{\sqrt{1 + \frac{e^2 E_0^2 t^2}{2m^2 v_0^2}}}$ (4) $\frac{\lambda_0}{\sqrt{1 + \frac{e^2 E_0^2 t^2}{m^2 v_0^2}}}$

39. Two similar plano-convex lenses are combined together in three different ways as shown in the adjoining figure. The ratio of the focal lengths in three cases will be



- (1) $2 : 2 : 1$ (2) $1 : 1 : 1$
 (3) $1 : 2 : 2$ (4) $2 : 1 : 1$

40. In the nuclear decay given below:



the particles emitted in the sequence are:

- (1) γ, β, α (2) β, γ, α
 (3) α, β, γ (4) β, α, γ
41. A nucleus ${}^A_Z X$ has mass represented by $M(A, Z)$. If M_p and M_n denote the mass of proton and neutron respectively and BE the binding energy in MeV, then
- (1) $BE = M(A, z) - ZM_p - (A - Z)M_n$
 (2) $BE = [M(A, z) - ZM_p - (A - Z)M_n]c^2$
 (3) $BE = [zM_p + (A - Z)M_n - M(A, Z)]c^2$
 (4) $BE = [zM_p + AM_n - M(A, Z)]c^2$

42. The binding energy of nucleus is a measure of its

- (1) Mass (2) Stability
- (3) Charge (4) Momentum

43. The barrier potential of p - n junction depends on:

- (A) Type of semi conductor material
- (B) Amount of doping
- (C) Temperature

Which one of the following is **correct**?

- (1) (A) and (B) only
- (2) (B) only
- (3) (B) and (C) only
- (4) (A), (B) and (C)

44. In forward biasing of the p - n junction

- (1) The positive terminal of the battery is connected to p -side and the depletion region becomes thin
- (2) The positive terminal of the battery is connected to p -side and the depletion region becomes thick
- (3) The positive terminal of the battery is connected to n -side and the depletion region becomes thin
- (4) The positive terminal of the battery is connected n -side and the depletion region becomes thick

45. Which one of the following gates correspond to the truth table given below?

A	B	Y
1	1	0
1	0	1
0	1	1
0	0	1

- (1) NAND (2) NOR
- (3) XOR (4) OR

46. Nature of image of a real object formed by convex mirror is always

- (1) Virtual, erect and magnified
- (2) Virtual, erect and diminished
- (3) Virtual, inverted and diminished
- (4) Real, inverted and diminished

47. **Assertion (A):** The resistivity of a semiconductor increases with temperature.

Reason (R): The atoms of a conductor vibrate with larger amplitude at higher temperature there by increasing it's resistivity.

- (1) Both **Assertion (A)** and **Reason (R)** are the true, and **Reason (R)** is a correct explanation of **Assertion (A)**.
- (2) Both **Assertion (A)** and **Reason (R)** are the true, but **Reason (R)** is not a correct explanation of **Assertion (A)**.
- (3) **Assertion (A)** is true, and **Reason (R)** is false.
- (4) **Assertion (A)** is false, and **Reason (R)** is true.

48. As the mass number A increases, the binding energy per nucleon in a nucleus-

- (1) Increases
- (2) Decreases
- (3) Remains the same
- (4) Varies in a way that depends on the actual value of A

49. The image of a virtual object formed by a concave mirror

- (1) May be virtual
- (2) May be real
- (3) Is always real
- (4) Is always virtual

50. **Assertion (A):** An N -type semiconductor has a large number of electrons but still it is electrically neutral.

Reason (R): An N -type semiconductor is obtained by doping an intrinsic semiconductor with a trivalent impurity.

- (1) Both **Assertion (A)** and **Reason (R)** are the true, and **Reason (R)** is a correct explanation of **Assertion (A)**.
- (2) Both **Assertion (A)** and **Reason (R)** are the true, but **Reason (R)** is not a correct explanation of **Assertion (A)**.
- (3) **Assertion (A)** is true, and **Reason (R)** is false.
- (4) **Assertion (A)** is false, and **Reason (R)** is true.

SECTION-(II) CHEMISTRY

SECTION - A

51. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

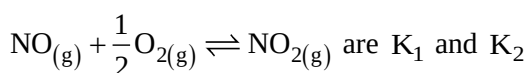
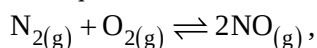
Assertion A: For any chemical reaction at a particular temperature, the equilibrium constant is fixed and is a characteristic property.

Reason R: Equilibrium constant is independent of temperature.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

52. The equilibrium constant for the reactions,



respectively. Then the equilibrium constant (K_c) for the equilibrium $\text{N}_{2(g)} + 2\text{O}_{2(g)} \rightleftharpoons 2\text{NO}_{2(g)}$

is:

- (1) K_1 / K_2
 - (2) $K_1 K_2$
 - (3) $K_1 K_2^2$
 - (4) $K_1^2 K_2$
53. The pH of a solution is 5. To this solution acid was added so that its pH value becomes 2.0, the increase in H^+ concentration is:
- (1) 100 times.
 - (2) 5 times.
 - (3) 2.5 times.
 - (4) 1000 times.

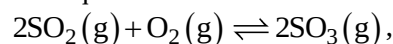
54. In which of the following equilibria, the value of K_p is less than K_c :

- (1) $\text{H}_2(g) + \text{I}_2(g) \rightleftharpoons 2\text{HI}(g)$
- (2) $\text{N}_2(g) + 3\text{H}_2(g) \rightleftharpoons 2\text{NH}_3(g)$
- (3) $\text{N}_2(g) + \text{O}_2(g) \rightleftharpoons 2\text{NO}(g)$
- (4) $\text{CO}(g) + \text{H}_2\text{O}(g) \rightleftharpoons \text{CO}_2 + \text{H}_2(g)$

55. The solubility of a sparingly soluble salt AB_2 in water is $1.0 \times 10^{-5} \text{ mol L}^{-1}$. Its solubility product of AB_2 will be:

- (1) 4×10^{-15}
- (2) 4×10^{-10}
- (3) 1×10^{-15}
- (4) 1×10^{-10}

56. The equilibrium:



shifts in forward direction if:

- (1) catalyst is added.
- (2) large amount of product is added.
- (3) small amounts of reactants are removed.
- (4) an adsorbent is used to remove $\text{SO}_3(g)$ as soon as it is formed.

57. Which will make basic buffer?

- (1) 50mL of 0.1M NaOH + 25mL of 0.1M CH_3COOH
- (2) 100mL of 0.1M CH_3COOH + 100mL of 0.1M NaOH
- (3) 100mL of 0.1M HCl + 200mL of 0.1M NH_4OH
- (4) 100mL of 0.1M HCl + 100mL of 0.1M NaOH

58. Energy of an electron is given by $E = -2.178 \times 10^{-18} \text{ J} \left(\frac{Z^2}{n^2} \right)$. Wavelength of light

required to excite an electron in a hydrogen atom from level $n = 1$ to $n = 2$ will be:

($h = 6.62 \times 10^{-34} \text{ Js}$ and $c = 3.0 \times 10^8 \text{ ms}^{-1}$)

- (1) $1.214 \times 10^{-7} \text{ m}$
- (2) $2.816 \times 10^{-7} \text{ m}$
- (3) $6.500 \times 10^{-7} \text{ m}$
- (4) $8.500 \times 10^{-7} \text{ m}$

59. If (p) is the momentum of the fastest electron ejected from a metal surface after the irradiation of light having wavelength λ , then for 1.5(p) momentum of the photoelectron, the wavelength of the light should be: (Assume kinetic energy of ejected photoelectron to be very high in comparison to work function)

- (1) $\frac{1}{2}\lambda$
- (2) $\frac{3}{4}\lambda$
- (3) $\frac{2}{3}\lambda$
- (4) $\frac{4}{9}\lambda$

60. The kinetic energy of an electron in the second Bohr orbit of a hydrogen atom is:

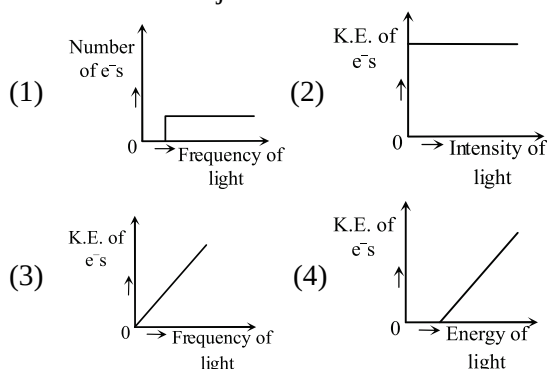
(where, a_0 is Bohr radius)

- (1) $\frac{h^2}{4\pi^2 m a_0^2}$
- (2) $\frac{h^2}{16\pi^2 m a_0^2}$
- (3) $\frac{h^2}{32\pi^2 m a_0^2}$
- (4) $\frac{h^2}{64\pi^2 m a_0^2}$

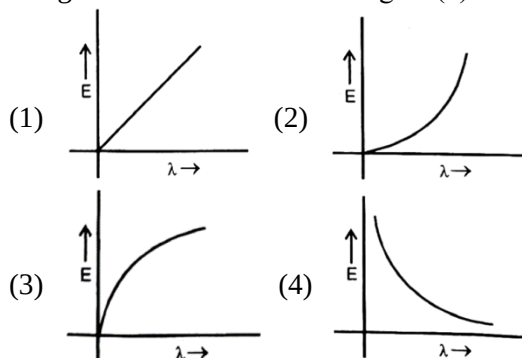
61. For emission line of atomic hydrogen from $n_i = n$ to $n_f = 8$, the plot of wave number ($\bar{\nu}$) against $\left(\frac{1}{n^2}\right)$ will be: (The Rydberg constant, R_H is in wave number unit)

- (1) Linear with slope - R_H
- (2) Linear with intercept - R_H
- (3) Non-linear
- (4) Linear with slope R_H

62. Which of the graphs shown below does **not** represent the relationship between incident light and the electron ejected from metal surface?



63. Which graph shows how the energy (E) of a photon of light is related to its wavelengths (λ)?



64. Which of the following species has maximum charge to mass ratio?

- (1) D^+
- (2) H^+
- (3) He^+
- (4) He^{2+}

65. Which of the following set of quantum numbers is possible for a particular electron in an atom is:

- (1) $n = 3, l = 0, m = 0, s = +\frac{1}{2}$
- (2) $n = 2, l = 2, m = 1, s = +\frac{1}{2}$
- (3) $n = 1, l = 0, m = -1, s = -\frac{1}{2}$
- (4) $n = 3, l = 2, m = 3, s = +\frac{1}{2}$

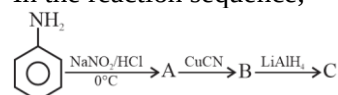
66. Match List-I with List-II:

List-I		List-II	
(A)	Number of values of 'l' for an energy level (n)	(I)	0, 1, 2, ..., (n - 1)
(B)	Values of 'l' for a particular type of orbital	(II)	-l to +l including zero
(C)	Number of value of 'm' for l = 2	(III)	5
(D)	Values of 'm' for a particular type of orbital	(IV)	n

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-I, D-II
- (2) A-IV, B-I, C-II, D-III
- (3) A-IV, B-I, C-III, D-II
- (4) A-I, B-IV, C-III, D-II

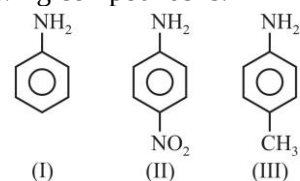
67. In the reaction sequence;



The product C is:

- (1) Benzonitrile
- (2) Benzaldehyde
- (3) Benzoic acid
- (4) Benzyl amine

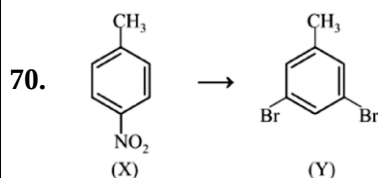
68. The **correct** increasing order of basic strength for the following compounds is:



- (1) (II) < (III) < (I)
- (2) (III) < (I) < (II)
- (3) (III) < (II) < (I)
- (4) (II) < (I) < (III)

69. The amine that reacts with Hinsberg's reagent to give an alkali insoluble product is:

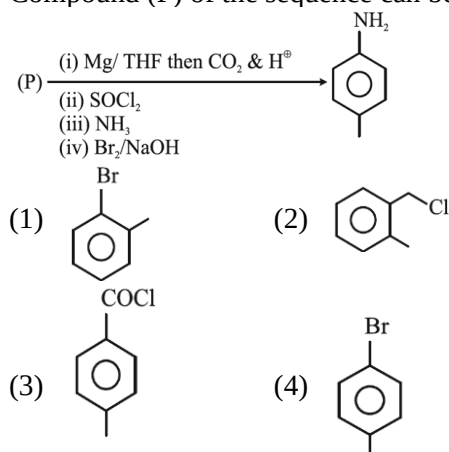
- (1) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{NH}-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_3$
- (2) $\text{CH}_3-\text{CH}_2-\underset{\text{CH}_2\text{CH}_3}{\text{N}}-\text{CH}_2\text{CH}_3$
- (3) $\text{CH}_3-\underset{\text{CH}_3}{\text{C}}(\text{NH}_2)-\text{CH}_2\text{CH}_2\text{CH}_3$
- (4) $\text{CH}_3-\underset{\text{CH}_3}{\text{C}}(\text{CH}_3)-\underset{\text{CH}_3}{\text{CH}}-\text{NH}_2$



In the above conversion of compound X to product Y, the sequence of reagents to be used will be:

- (1) (i) Br_2, Fe (ii) Fe, H^+ (iii) LiAlH_4
- (2) (i) Br_2 (aq) (ii) LiAlH_4 (iii) H_3O^+
- (3) (i) Fe, H^+ (ii) Br_2 (aq) (iii) HNO_2 (iv) CuBr
- (4) (i) Fe, H^+ (ii) Br_2 (aq) (iii) HNO_2 (iv) H_3PO_2

71. Compound (P) of the sequence can be:



72. Given below are two statements:

Statement I: Aniline is less basic than acetamide.

Statement II: In aniline, the lone pair of electrons on nitrogen atom is delocalised over benzene ring due to resonance and hence less available for protonation.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

73. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

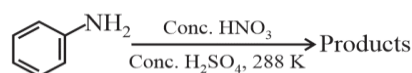
Assertion A: Amines are more basic than alcohols.

Reason R: Amines can readily donate a lone pair of electrons, while alcohols cannot do so as effectively.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

74. With respect to the following reaction, consider the given statements:

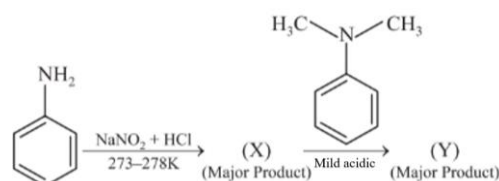


- I. o-Nitroaniline and p-nitroaniline are the predominant products.
- II. p-Nitroaniline and m-nitroaniline are the predominant products.
- III. HNO_3 acts as an acid.
- IV. H_2SO_4 acts as an acid.

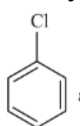
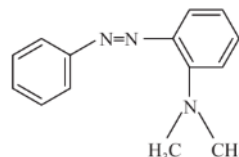
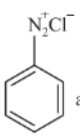
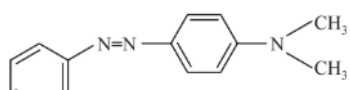
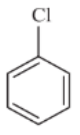
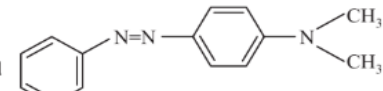
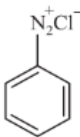
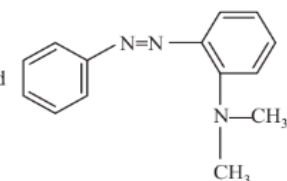
Choose the **correct** option.

- (1) I and III only (2) I and IV only
- (3) II and IV only (4) II and III only

75.



Considering the above reaction, X and Y respectively are:

- (1)  and 
- (2)  and 
- (3)  and 
- (4)  and 

76. Match List-I with List-II:

List-I (Vitamins)		List-II (Deficiency disease)	
(A)	Vitamin A	(I)	Scurvy
(B)	Vitamin B ₁₂	(II)	Pernicious anaemia
(C)	Vitamin C	(III)	Muscular weakness
(D)	Vitamin E	(IV)	Xerophthalmia

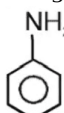
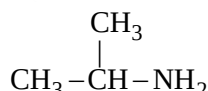
Choose the **correct** answer from the options given below:

- (1) A-II, B-IV, C-I, D-III
- (2) A-IV, B-III, C-I, D-II
- (3) A-IV, B-II, C-I, D-III
- (4) A-III, B-IV, C-I, D-II

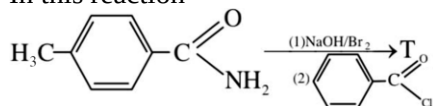
77. Given below are two statements:
Statement I: Schiff's test is used in the detection of aldehyde functional group.
Statement II: Sodium hydrogencarbonate test is used for the detection of carboxylic acid functional group.
 In the light of the above statements, choose the *most appropriate* answer from the options given below:
- Statement I is correct but Statement II is incorrect.
 - Statement I is incorrect but Statement II is correct.
 - Both Statement I and Statement II are correct.
 - Both Statement I and Statement II are incorrect.
78. In 3rd group cationic analysis, NH_4Cl is added before adding NH_4OH . This is due to:
- decrease the concentration of OH^- .
 - increase the concentration of OH^- .
 - increase concentration of Cl^- .
 - prevent interference of PO_4^{3-} ion.
79. A gas X is added to Nessler's reagent then brown precipitate is formed. The gas 'X' is:
- Cl_2
 - Br_2
 - SO_2
 - NH_3
80. Which of the following ion pair can be separated by using dilute HCl ?
- Ag^+ and Cu^{2+}
 - Ag^+ and Al^{3+}
 - Hg_2^{2+} and Cd^{2+}
 - All of these
81. A solution containing a group IV cation gives a precipitate on passing H_2S . A solution of this precipitate in dil. HCl produces a white precipitate with NaOH solution and a bluish-white precipitate with basic potassium ferrocyanide. The cation is:
- Co^{2+}
 - Ni^{2+}
 - Mn^{2+}
 - Zn^{2+}
82. An inorganic salt is strongly heated the residue is yellow when hot and white when, cold. The salt contains:
- Cu^{2+}
 - Zn^{2+}
 - Co^{2+}
 - Fe^{2+}
83. Na_2CO_3 **cannot** be used in place of $(\text{NH}_4)_2\text{CO}_3$ for the precipitation of V group because:
- Na^+ interferes in the detection of V group.
 - conc. of CO_3^{2-} is very low.
 - Na will react with acid radicals.
 - Mg will be precipitated.

84. An ammoniacal metal salt solution gives a brilliant red precipitate on addition of dimethylglyoxime. The metal is:
- Ni^{2+}
 - Cu^{2+}
 - Fe^{2+}
 - Co^{2+}
85. A gas is obtained by addition of dil. H_2SO_4 to a mixture which turns lead acetate paper black. It is:
- SO_2
 - CO_2
 - H_2S
 - NO_2

SECTION - B

86. For ortho phosphoric acid,
 $\text{H}_3\text{PO}_4(\text{aq}) + \text{H}_2\text{O}(\text{aq}) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{H}_2\text{PO}_4^-(\text{aq}); K_{a_1}$
 $\text{H}_2\text{PO}_4^-(\text{aq}) + \text{H}_2\text{O}(\text{aq}) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{HPO}_4^{2-}(\text{aq}); K_{a_2}$
 $\text{HPO}_4^{2-}(\text{aq}) + \text{H}_2\text{O}(\text{aq}) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{PO}_4^{3-}(\text{aq}); K_{a_3}$
 The **correct** order of K_{a_1} , K_{a_2} and K_{a_3} values is:
- $K_{a_1} > K_{a_2} > K_{a_3}$
 - $K_{a_1} < K_{a_2} < K_{a_3}$
 - $K_{a_1} > K_{a_3} > K_{a_2}$
 - $K_{a_1} > K_{a_2} = K_{a_3}$
87. If $K_a = 10^{-6}$ for weak acid, then $\text{p}K_b$ of its conjugate base is:
- 10^{-8}
 - 8
 - 6
 - 10^{-7}
88. The addition of sodium acetate to acetic acid solution will cause:
- increase in its pH value.
 - decrease in its pH value.
 - no change in pH value.
 - None of these
89. Which of the following reactions proceed at low pressure?
- $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$
 - $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$
 - $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$
 - $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$
90. Which of the following amines will **not** give positive test with $\text{CHCl}_3 / \text{KOH}(\text{alc.})$?
- $\text{CH}_3-\text{CH}_2-\text{NH}-\text{CH}_3$
 - $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{NH}_2$
 - 
 - 

91. In this reaction



The structure of T would be:

- (1)
- (2)
- (3)
- (4)

92. The amide which does **not** undergo Hoffmann's bromamide reaction is:

- (1) acetamide. (2) benzamide.
(3) formamide. (4) propanamide.

93. Benzene diazonium chloride on reaction with boiling water gives:

- (1) benzylamine. (2) benzaldehyde.
(3) aniline. (4) phenol.

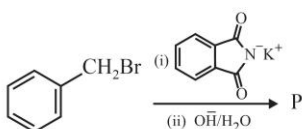
94. The **correct** statement regarding the basicity of arylamine is:

- (1) arylamines are generally less basic than alkylamine because the nitrogen lone-pair electrons are delocalized by interaction with the aromatic ring π -electron system.
(2) arylamines are generally more basic than alkylamine because the nitrogen lone pair electrons are delocalized by interaction with the aromatic ring π -electron system.
(3) arylamines are generally more basic than alkylamine because the nitrogen lone pair electrons are not delocalized by interaction with the aromatic ring π -electron system.
(4) arylamines are generally more basic than alkylamine because the nitrogen atom in arylamines sp -hybridized.

95. Which of the following **cannot** be prepared by Sandmeyer's reaction?

- I. Chlorobenzene II. Bromobenzene
III. Iodobenzene IV. Fluorobenzene
(1) I and II only (2) II and III only
(3) III and IV only (4) I and IV only

96. The major product (P) formed in the following reaction is:



- (1)
- (2)
- (3)
- (4)

97. Glucose on prolonged heating with HI gives:

- (1) 1-Hexene
(2) Hexanoic acid
(3) 6-Iodohexanal
(4) n-Hexane

98. The **correct** statement regarding RNA and DNA, respectively is:

- (1) the sugar component in RNA is β -D-arabinose and the sugar component in DNA is β -D-ribose.
(2) the sugar component in RNA is β -D-2-deoxyribose and the sugar component in DNA is β -D-arabinose.
(3) the sugar component in RNA is β -D-arabinose and the sugar component in DNA is β -D-2-deoxyribose.
(4) the sugar component in RNA is β -D-ribose and the sugar component in DNA is β -D-2-deoxyribose.

99. The glycosidic linkage present in sucrose is between:

- (1) C - 1 of α -glucose and C - 2 of β -fructose.
(2) C - 1 of α -galactose and C - 4 of α -glucose.
(3) C - 1 of α -glucose and C - 4 of α -glucose.
(4) C - 1 of α -glucose and C - 4 of β -fructose.

100. Match List-I with List-II:

List-I (Species)		List-II (Conjugate acid)	
(A)	NH_3	(I)	H_2CO_3
(B)	HCO_3^-	(II)	NH_4^+
(C)	H_2O	(III)	H_3O^+
(D)	HSO_4^-	(IV)	H_2SO_4

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-II, D-IV
(2) A-II, B-I, C-III, D-IV
(3) A-IV, B-III, C-II, D-I
(4) A-II, B-III, C-I, D-IV

SECTION-(III) BOTANY**SECTION - A**

101. Which of the following is **correct** about photosynthetically active radiation (PAR)?

- (1) It is the amount of sunlight that is not required for the photosynthesis.
- (2) It is the part of sunlight that is utilised by plants for photosynthesis.
- (3) Less than 2% of PAR is captured by plants.
- (4) More than 50% of solar radiation is PAR.

102. Match **List-I** with **List-II**.

List-I (Extinct animals)		List-II (Country)	
A.	Dodo	I.	Africa
B.	Quagga	II.	Russia
C.	Thylacine	III.	Mauritius
D.	Steller's Sea Cow	IV.	Australia

Choose the **correct** answer from the option given below:

- (1) A-II, B-III, C-I, D-IV
- (2) A-II, B-III, C-IV, D-I
- (3) A-III, B-I, C-IV, D-II
- (4) A-III, B-II, C-IV, D-I

103. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Net primary productivity is gross primary productivity minus respiration.

Reason R: Secondary productivity is produced by heterotrophs.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

104. Out of the total annual net primary productivity of the whole biosphere, the productivity of the oceans is:

- (1) approximately 170 billion tons (dry weight) of organic matter.
- (2) only 55 billion tons (dry weight) of organic matter.
- (3) 115 billion tons (dry weight) of organic matter.
- (4) only 70 billion tons (dry weight) of organic matter.

105. The Earth summit held in Rio de Janeiro in 1992 was called:

- (1) to reduce CO₂ emissions and global warming.
- (2) for conservation of biodiversity and sustainable utilisation of its benefits.
- (3) to assess threat posed to native species by invasive weed species.
- (4) for immediate steps to discontinue use of CFCs that were damaging the ozone layer.

106. In *lac* operon:

- (1) promoter gene interacts with regulatory protein.
- (2) regulatory gene possesses the site for RNA polymerase attachment.
- (3) *Lac z* codes for an enzyme that increases permeability of cell to β -galactosides.
- (4) the *i* gene codes for repressor protein.

107. Given below are two statements:

Statement I: In an aquatic ecosystem, GFC is the major conduit for energy flow.

Statement II: In a terrestrial ecosystem, a much larger fraction of energy flows through the detritus food chain.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

108. $\frac{dN}{dt} = rN \left(\frac{K - N}{K} \right)$ in logistic growth K represents:

- (1) carrying capacity.
- (2) exponential factor.
- (3) intrinsic rate of natural increases.
- (4) intrinsic rate of natural decreases.

109. Dead plant remains such as leaves, bark, flowers and dead remains of animals, including fecal matter, constitute detritus, which is the raw material for _____.

- (1) decomposition (2) leaching
(3) humification (4) mineralisation

110. The rate of decomposition is faster in the ecosystem due to following factors **except**:

- (1) warm and moist environment.
(2) presence of aerobic soil microbes.
(3) detritus richer in lignin and chitin.
(4) detritus rich in sugars.

111. Match **List-I** with **List-II**.

List I		List II	
A.	The Evil Quartet	I.	Rivet popper hypothesis
B.	Paul Ehrlich	II.	Presence of exclusive species in an area
C.	Endemism	III.	Direct economic benefits from nature
D.	Narrowly utilitarian	IV.	Four major causes of biodiversity losses

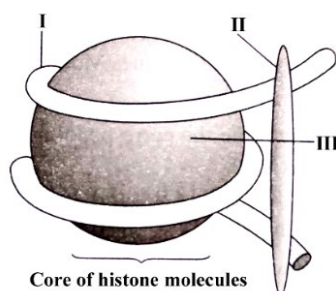
Choose the **correct** answer from the option given below:

- (1) A-III, B-II, C-IV, D-I
(2) A-II, B-III, C-IV, D-I
(3) A-I, B-III, C-II, D-IV
(4) A-IV, B-I, C-II, D-III

112. Each trophic level has certain mass of living material at a particular time called as:

- (1) standing crop
(2) net primary productivity (NPP).
(3) standing state.
(4) gross primary productivity (GPP).

113. The given figure shows the structure of nucleosome with their parts labelled as **I**, **II** & **III**. Identify **I**, **II** and **III**.



- (1) I - DNA; II - H1 histone; III - Histone octamer
(2) I - H1 Histone; II - DNA; III - Histone octamer
(3) I - Histone octamer; II - RNA; III - H1 histone
(4) I - RNA; II - H1 histone; III - Histone octamer

114. Identify the **incorrect** statement.

- (1) Depending on base composition (A:T rich or G:C rich), length of segment, and number of repetitive units, satellite DNA is classified into micro-satellites and mini-satellites etc.
(2) The technique of DNA fingerprinting was initially developed by Alec Jeffreys.
(3) Polymorphisms are non-inheritable from parents to children.
(4) DNA fingerprinting is the basis of paternity testing, in case of disputes.

115. Population growth of a species is **not** related to which of the following parameters?

- (1) $N_{t+1} = N_t + [(B + I) - (D + E)]$
(2) Out of the four basic processes affecting population growth, three lead to increase in population size.
(3) For a newly colonized habitat, immigration plays major role than the birth rate.
(4) Under normal conditions, natality and mortality are the most important factors.

116. Give right answer for the following:

- A. Number of births during a given period in the population that are added to initial density.
B. Number of deaths in the population in a given period.
C. Number of individuals of the same species that have come into the habitat from elsewhere during a given time period.
D. Number of individuals who left habitat and gone elsewhere during a given time period.
(1) A-mortality, B-natality, C-emigration, D-immigration
(2) A- mortality, B- natality, D-emigration, C- immigration
(3) B-mortality, A-natality, C- emigration, D- immigration
(4) B- mortality, A natality-, D- emigration, C- immigration

117. Which of the following pairs of organisms does **not** show '+, -' relationship?
- (1) Ticks and dogs
 - (2) Lice and humans
 - (3) *Cuscuta* and hedge plants
 - (4) Cattle egret and grazing cattle

118. Mark the **incorrect** statement regarding biodiversity hotspots.

- (1) A very high level of species richness, species confined to that region and not found anywhere else
- (2) The 34 biodiversity hotspots cover almost 30% of earth's land surface
- (3) These hotspots are also region of accelerated habitat loss
- (4) Western ghats in India is a biodiversity hotspot

119. Match List-I with List-II.

List-I		List-II	
A.	Mutualism	I.	Tiger and deer
B.	Commensalism	II.	Copepods and marine fish
C.	Parasitism	III.	Fig and its pollinator
D.	Predation	IV.	Orchid and mango tree

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III, D-IV
- (2) A-III, B-IV, C-II, D-I
- (3) A-II, B-III, C-I, D-IV
- (4) A-IV, B-II, C-III, D-I

120. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Some amino acids are coded by more than one codon.

Reason R: The codon is read in tRNA in a contiguous fashion.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

121. Although India has only ____ **P** ____ percent of the world's land area, its share of the global species diversity is an impressive ____ **Q** ____ percent. Identify **P** and **Q** respectively.

- (1) **P**-3.4, **Q**-4.8 (2) **P**-1.4, **Q**-6.8
- (3) **P**-2.4, **Q**-8.1 (4) **P**-8.1, **Q**-9.2

122. Western Ghats have a large number of plant and animal species that are **not** found anywhere else. Which of the following terms will you use to notify such species?

- (1) Endemic
- (2) Vulnerable
- (3) Threatened
- (4) Keystone

123. A decline in population size will be seen when:

- (1) Natality > Mortality
- (2) Immigration < Emigration
- (3) Immigration = Emigration
- (4) Both (1) and (2)

124. MacArthur showed that five closely related species of warblers living on the same tree were able to avoid competition and co-exist due to behavioural differences in:

- (1) high reproductive capacity.
- (2) intrinsic rate of increase is very low.
- (3) their foraging activities.
- (4) competitive release.

125. Read the following statements and select the **correct** option w.r.t. population attributes.

- A. Population density is not necessarily measured in numbers.
- B. Tiger census is often based on pug marks and fecal pellets.
- C. Biomass is not a meaningful measure to know population size.
- D. Relative density is a good measure to find out the density of fishes in the lake.

Choose the most appropriate answer from the options given below:

- (1) Only A
- (2) Only A & B
- (3) All except C
- (4) All, A, B, C & D

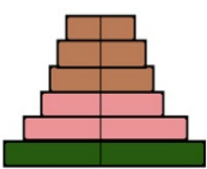
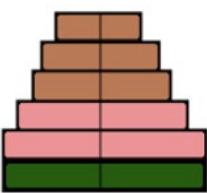
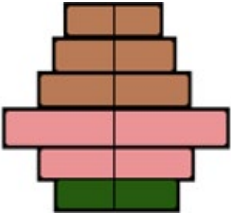
126. What would be happen if both the strands of DNA are copied during transcription?

- (1) One segment of DNA would be coding for two different proteins.
- (2) Two RNA will be produced simultaneously, complementing to each other.
- (3) Formation of double helical RNA.
- (4) All of these

127. Which of the following statement is **correct**?

- (1) Warm and moist environment favour decomposition where as low temperature and anaerobic inhibit decomposition.
- (2) Warm and moist environment inhibit decomposition whereas low temperature and anaerobiosis favor decomposition.
- (3) Warm moist environment and anaerobiosis inhibit decomposition whereas low temperature favours decomposition.
- (4) Warm and low temperature inhibit decomposition whereas anaerobic favours decomposition.

128. A population in which number of pre-reproductive individuals is very large, number of reproductive individuals is moderate while post-reproductive individuals are fewer show which kind of age pyramid?

- (1) 
- (2) 
- (3) 
- (4) Cannot be determined

129. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Competition occurs when closely related species compete for the same resources that are abundant.

Reason R: Interspecific competition is a potent force in organic evolution.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

130. According to Alexander von Humboldt:

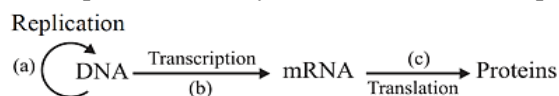
- (1) species richness increases with increasing area, but only up to limit.
- (2) there is no relationship between species richness and area explored.
- (3) species richness goes on increasing with increasing area of exploration.
- (4) species richness decreases with increasing area of exploration.

131. $\log S = \log C + Z \log A$

Above equation is derived from the relationship between species richness and area, on a logarithmic scale, which is a straight line. Value of Z if we analyse in very large areas (like continents) will be between:

- (1) 0.1 – 0.2.
- (2) 0.6 – 1.2.
- (3) 2 – 4.
- (4) 5 – 10.

132. In the given flow chart of central dogma, select the **correct** option w.r.t. enzymes involved in the steps.



- (1) (a) DNA polymerase
(b) RNA polymerase
(c) Ribozyme
- (2) (a) RNA polymerase
(b) Reverse transcriptase
(c) DNA polymerase
- (3) (a) DNA dependent DNA polymerase
(b) RNA dependent DNA polymerase
(c) Ribozyme
- (4) (a) RNA dependent RNA polymerase
(b) Transcriptase
(c) DNA polymerase

133. Which of the following is **not** a characteristic of humus?

- (1) It is rich in organic matter such as lignin and cellulose.
- (2) It is colloidal in nature and serves as a reservoir of nutrients.
- (3) It is highly resistant to microbial action and undergoes slow decomposition.
- (4) It is further degraded by the process of humification.

134. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: The hnRNA in humans has exons and introns.

Reason R: The primary transcript produced in eukaryotes is translated without undergoing any modification or processing.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

135. Amensalism is an association between two species where:

- (1) one species is harmed and other is benefitted.
- (2) one species is harmed and other is unaffected.
- (3) one species is benefitted and other is unaffected.
- (4) both the species are harmed.

SECTION - B

136. What would happen if in a gene encoding a polypeptide of 50 amino acids, 25th codon (UAU) is mutated to UAA?

- (1) A polypeptide of 25 amino acids will be formed.
- (2) A polypeptide of 24 amino acids will be formed.
- (3) Two polypeptides of 24 and 25 amino acids will be formed.
- (4) A polypeptide of 49 amino acids will be formed.

137. The pyramid of biomass in sea is generally A because the biomass of fishes far B that of phytoplankton. Identify A and B.

- (1) A-upright, B-less
- (2) A-upright, B-exceed
- (3) A-inverted, B-less
- (4) A-inverted, B-exceed

138. Amazon rain forest is being cut and cleared for:

- (1) cultivating soyabeans only.
- (2) for conversion to grasslands for raising beef cattle only.
- (3) for making roads.
- (4) Both (1) and (2)

139. Choose **incorrect** statement.

- (1) Split gene arrangement in eukaryotes represents an ancient feature of genome.
- (2) The presence of introns is reminiscent of antiquity.
- (3) Process of splicing represents the dominance of RNA-world.
- (4) Splicing represent the dominance of DNA world.

140. In a field experiment, when all *Pisaster* starfish were removed from an enclosed intertidal area, the result was:

- (1) increase in diversity of invertebrates.
- (2) extinction of many invertebrate species.
- (3) inability of the *Pisaster* to enter the area again.
- (4) replacement of *Pisaster* by other starfish.

141. Mostly in eukaryotes, the structural genes are said to be monocistronic:

- (1) have only one intron.
- (2) have only one exon.
- (3) have codons that code for only one type of amino acid.
- (4) have interrupted coding sequences.

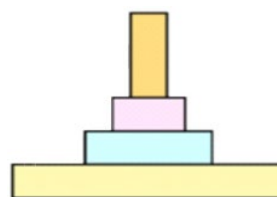
142. Match List-I with List-II.

List-I		List-II	
A.	David Tilman	I.	Long-term ecosystem experiment
B.	Communities with more species	II.	Term biodiversity
C.	Communities with less species	III.	Less stable
D.	Edward Wilson	IV.	More stable

Choose the **correct** answer from the option given below:

- (1) A-II, B-IV, C-III, D-I
- (2) A-I, B-IV, C-III, D-II
- (3) A-I, B-III, C-IV, D-II
- (4) A-I, B-IV, C-II, D-III

143. Select the most suitable option for the diagram given below.



- (1) Pyramid of numbers in a grassland ecosystem
- (2) Pyramid of biomass in a pond ecosystem
- (3) Pyramid of numbers in a tree ecosystem
- (4) Both (1) and (2)

144. Genetic code:
- (1) could direct the sequence of amino acids during synthesis of proteins.
 - (2) triplet base on mRNA.
 - (3) is nearly universal.
 - (4) All of these

145. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Colombia located near the equator has nearly 1,400 species of birds while New York at 41° N has 105 species.

Reason R: In general, species diversity decreases as we move away from the equator towards the poles.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

146. Gause's 'Competitive Exclusion Principle' states that:

- (1) more abundant species will exclude the less abundant species through competition.
- (2) competition for the same resources excludes species having different food preferences.
- (3) two closely related species competing for the same resources cannot co-exist indefinitely for the same limiting resources.
- (4) larger organisms exclude smaller ones through competition.

147. Match List-I with List-II.

List-I		List-II	
A.	Genetic diversity	I.	Co-extinctions
B.	Species diversity	II.	Red list(2004)
C.	IUCN	III.	Amphibian species
D.	Cause of biodiversity loss	IV.	<i>Rauwolfia vomitoria</i>

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
- (2) A-III, B-I, C-II, D-IV
- (3) A-III, B-II, C-I, D-IV
- (4) A-III, B-II, C-IV, D-I

148. Read the following statements given below:

- A. Predators maintain species diversity.
- B. Ecologically, herbivores are predators.
- C. Interspecific competition is more intense than intraspecific competition.
- D. Resource may not be limiting for competition to occur as in case of interference competition.

Choose the most appropriate answer from the options given below:

- (1) Only A, C and D are correct
- (2) Only A, B and D are correct
- (3) Only A, B and C are correct
- (4) Only B, C and D are correct

149. Read the following statements and select the **correct** ones.

- A. At present, the world is facing the sixth mass extinction of species.
- B. 'Rivet popper hypothesis' explains that key species are least important.
- C. Species area relationship was explained by Alexander von Humboldt.
- D. Species diversity is more in Eastern Ghats than Western Ghats.
- E. According to the (IUCN) (2004), the total number of plant and animal species described so far is slightly more than 1.5 million.

Choose the most appropriate answer from the options given below:

- (1) A, C and D
- (2) A, B and C
- (3) A, C and E
- (4) A, B and D

150. Which of the following is **not** a salient feature of the Human Genome Project?

- (1) The human genome contains 3614.7 million nucleotide bases.
- (2) Less than 2 per cent genome codes for proteins.
- (3) Over 50 per cent of the genes discovered are yet to be investigated regarding their functions.
- (4) Scientists have identified about 1.4 million locations where single base DNA differences (SNPs – single nucleotide polymorphism, pronounced as 'snips') occur in humans.

SECTION-(IV) ZOOLOGY**SECTION - A**

151. How many recombinant therapeutics have been approved for human use the world over?

- (1) 30 (2) 18
(3) 50 (4) 12

152. How many from the following crops are now genetically modified by *Bt*-toxin gene?

Cotton, Corn, Rice, Tomato, Potato, Soyabean
--

- (1) Four (2) Five
(3) Three (4) Six

153. Given below are two statements:

Statement I: Nematode-specific genes were introduced into the host plant by using *Agrobacterium* vector.

Statement II: The capacity to generate a whole plant from any cell or explant is called totipotency. In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

154. What is a concern regarding patents granted for traditional herbal medicines.

- (1) They may lead to loss of rights for local communities.
(2) They have no impact on traditional practices.
(3) They are easily enforced.
(4) They benefit the original communities.

155. Which of the following was the first to be used as the source of biopesticide on the commercial scale in the world?

- (1) *Bacillus thuringiensis*
(2) *Escherichia coli*
(3) *Pseudomonas aeruginosa*
(4) *Agrobacterium tumefaciens*

156. Read the following statements (A-E).

- A. The green revolution succeeded in tripling the food supply.
B. Transgenic fish are animals that have had their DNA manipulated to possess and express an extra foreign gene.
C. There are an estimated only 2000 rice varieties in India.

D. 95 per cent of all existing transgenic animals are mice.

E. Mature insulin consists of three polypeptide chains

How many of the above statements is/are **correct**?

- (1) Two
(2) One
(3) Four
(4) Three

157. Genetic modification is being used to create tailor-made plants in order to supply alternative resources to industries in the form of:

- (1) starch.
(2) fuels.
(3) pharmaceuticals.
(4) all of these

158. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: *Bt* cotton contains a gene from *Bacillus thuringiensis* that protects it from viral infection.

Reason R: The *Bt* toxin gene produces an insecticidal protein that kills insects upon ingestion by disrupting their gut cells.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

159. What principle does PCR rely on?

- (1) DNA sequencing
(2) Protein synthesis
(3) Antigen-antibody interaction
(4) Amplification of DNA.

160. In _____ first clinical gene therapy administered _____.

Choose the option which fill the blank **correctly**.

- (1) 1980, to a 6-year-old boy with haemophilia
(2) 1990, to a 4-year-old girl with ADA deficiency
(3) 2000, to a 5-year-old boy with leukemia
(4) 2010, to a newborn with cystic fibrosis

161. The two polypeptides of human insulin are linked together by_____.

Choose the option which fill the blank **correctly**.

- (1) phosphodiester bonds.
- (2) ether bonds.
- (3) disulphide bridges.
- (4) hydrogen bonds.

162. Cotton bollworm are controlled by:

- (1) *cry* IAb. (2) *cry* IAc.
- (3) *cry* IIAb. (4) Both (2) and (3)

163. Which of the following is the first company to synthesise and market Human insulin?

- (1) Sun pharma
- (2) Torrent
- (3) Eli Lilly
- (4) Abbot laboratories

164. Match **List-I** with **List-II**:

List-I		List-II	
(A)	Emphysema	(I)	Micro propagation
(B)	Transgenic model	(II)	Can be treated by α -1 antitrypsin
(C)	Axillary meristem	(III)	Cystic fibrosis
(D)	Coleopterans	(IV)	Bettles

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
- (2) A-I, B-III, C-IV, D-II
- (3) A-I, B-II, C-III, D-IV
- (4) A-IV, B-III, C-II, D-I

165. Which of the following is a permanent cure for ADA deficiency?

- (1) Enzyme replacement therapy.
- (2) Bone marrow transplant.
- (3) Introduction of a functional ADA cDNA into lymphocytes.
- (4) Introduction of gene isolated from marrow cells producing ADA into early embryonic cells.

166. Match **List-I** with **List-II**:

List-I		List-II	
(A)	<i>Hind</i> II	(I)	Cellulase
(B)	Degradation of plant cell wall	(II)	Recognition sequence of six base pairs
(C)	Degradation of fungal cell wall	(III)	<i>tet</i> ^R
(D)	<i>pBR322</i>	(IV)	Chitinase

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-IV, D-I
- (2) A-III, B-I, C-II, D-IV
- (3) A-III, B-IV, C-I, D-II
- (4) A-II, B-I, C-IV, D-III

167. Which of the following diseases is an autoimmune disorder?

- (1) Tumor
- (2) Rheumatoid arthritis
- (3) Hepatitis
- (4) Gout

168. Given below are two statements.

Statement I: Computed tomography uses X-rays to generate a two-dimensional image.

Statement II: Traditional hybridisation procedures used in plant and animal breeding.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Statement I is correct, but Statement II is incorrect.
- (2) Statement I is incorrect, but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

169. The humoral immunity inside the human body is carried out by:

- (1) Antibodies.
- (2) T-lymphocytes.
- (3) Monocytes.
- (4) Thrombocytes.

170. Read the following statements (**A-D**).

- A. DNA fragments travel along the surface of the gel whose concentration does not affect movement of DNA.
- B. DNA is negatively charged and is loaded on the gel towards the anode terminal.
- C. Smaller the size of DNA fragment, shorter is the distance it travels through it.
- D. Pure DNA can be visualised only after staining the DNA with ethidium bromide.

Which of the following statements are **correct** regarding separation of DNA fragments using gel electrophoresis?

- (1) B and C only
- (2) C and D only
- (3) A and D only
- (4) D only

171. Which of the following vector is commonly used in the transfer of genes in a crop plant?

- (1) Plasmids of *Mycobacterium*
- (2) Bacteriophage
- (3) Ti-plasmids of *Agrobacterium*
- (4) *E.coli* phage

172. *Smack* is obtained from:

- (1) *Papaver somniferum*.
- (2) *Atropa belladonna*.
- (3) *Erythroxylum coca*.
- (4) *Datura*.

173. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: A thermostable DNA polymerase is isolated from a bacterium, *Agrobacterium*.

Reason R: *Taq* polymerase remains active during the high temperature.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R true and R is not the correct explanation of A

174. How many copies of DNA samples are produced in PCR technique after five cycles?

- (1) 4
- (2) 32
- (3) 64
- (4) 16

175. Which of the following enzymes is used to join DNA fragments?

- (1) Exonucleases
- (2) Endonuclease
- (3) Ligase
- (4) DNA polymerase

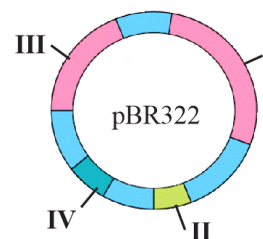
176. Given below are two statements.

Statement I: The normal *E.coli* carries resistance against both ampicillin and tetracycline.

Statement II: Stanley Cohen and Herbert Boyer isolating the first antibiotic resistance gene. In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Statement I is correct, but Statement II is incorrect.
- (2) Statement I is incorrect, but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

177. Identify the diagram given below and choose the **correct** option that represent *tet^R* selectable marker.



- (1) I
- (2) II
- (3) III
- (4) IV

178. In *EcoRI*, *co* stands for:

- (1) strain.
- (2) species.
- (3) genus.
- (4) order.

179. Read the following statements (A-E).

- A. Colostrum provides passive immunity.
- B. *ori* is responsible for inhibiting replication.
- C. 'MALAYALAM' is a palindrome sequence.
- D. Mosquitoes act as an insect vector to transfer the plague parasite into the human body.
- E. Acquired immunity is specific type of defense.

How many of the above statements are/are **incorrect**?

- (1) Two
- (2) One
- (3) Four
- (4) Three

180. Which of the following is **true** for cloning vectors?

- (1) More than two sites of origin of replication.
- (2) Vector should have selectable marker gene.
- (3) Single recognition site for the commonly used restriction enzyme.
- (4) Both (2) and (3)

181. An enzyme catalysing the removal of nucleotides from the ends of DNA is:

- (1) DNA ligase
- (2) Exonuclease
- (3) Endonuclease
- (4) DNA polymerase

182. HIV that causes AIDS first starts destroying:

- (1) B-lymphocytes.
- (2) leucocytes.
- (3) thrombocytes.
- (4) helper T-lymphocytes.

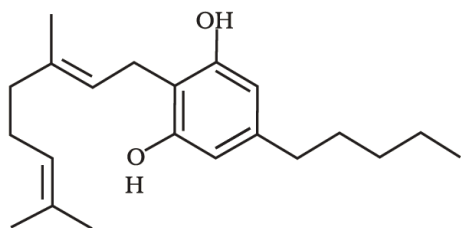
183. Which of the following is a secondary lymphoid organs of our body?

- (1) Bone marrow
- (2) Spleen
- (3) Lymph node
- (4) Both (2) and (3)

184. Which of the following is/are symptoms of withdrawal syndrome?

- (1) Anxiety (2) Shakiness
(3) Sweating (4) All of these

185. Identify the diagram given below and choose the **correct** one from the following given options.



- (1) Cannabinoid (2) Haemoglobin
(3) Lecithin (4) Morphine

SECTION - B

186. Read the following list carefully.

Skin, erythrocytes, monocytes, macrophages, NK cells, Tears

How many of the above cells constitute cellular barrier of innate immunity?

- (1) Six (2) Three
(3) Five (4) Two

187. Given below are two statements.

Statement I: *Cannabis* plants parts are used in various combination to produce marijuanas.

Statement II: The vaccines generate memory-B and T-cells that recognise the pathogen quickly on subsequent exposure.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Statement I is correct, but Statement II is incorrect.
(2) Statement I is incorrect, but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

188. Read the following statements (A-D).

- A. Heroin is depressant.
B. Crack has a potent stimulating action on the central nervous system.
C. Coke is obtained from *Erythroxylum coca*.
D. Smoking paves the way to hard drugs.

Which of the following statements are **correct**?

- (1) B and C only
(2) C and D only
(3) A, B and C only
(4) A, B, C, D only

189. Restriction enzyme *EcoRI* cuts the DNA between bases **G** and **A** only when the sequence in DNA is:

- (1) GATATC (2) GAATTC
(3) GATTCC (4) GAACTT

190. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: HIV-infection generally occurs by sexual contact with a non-infected person.

Reason R: There is always a time lag between the infection and appearance of AIDS symptoms. In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is not the correct explanation of A.

191. Read the following statements (A-E).

- A. Cocaine interferes with the transport of dopamine.
B. AIDS is an infectious disease.
C. Infectious diseases spread from unhealthy to healthy people.
D. Coke produces a sense of euphoria.
E. Healthy persons bring economic prosperity.

How many of the above statements is/are **correct**?

- (1) Two
(2) One
(3) Four
(4) Five

192. Separation and purification of the products that produced in bioreactor is a part of:

- (1) downstream processing.
(2) transcriptional processing.
(3) upstream processing.
(4) Both (2) and (3)

193. Match **List-I** with **List-II**:

List-I		List-II	
(A)	Acid in the stomach	(I)	Colostrum
(B)	IgE	(II)	Physiological barrier
(C)	IgA	(III)	Viral infection
(D)	HIV	(IV)	Allergy

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-IV, D-I
(2) A-II, B-IV, C-I, D-III
(3) A-III, B-IV, C-I, D-II
(4) A-III, B-I, C-IV, D-II

- 194.** Introduction of foreign genes for improving genotype is:
- (1) cell biology.
 - (2) tissue culture.
 - (3) vernalisation.
 - (4) genetic engineering.
- 195.** These are three basic steps in genetically modify organism arrange these steps in **correct** sequence
- I. Introduction of the Identified DNA into the host
 - II. Maintenance of introduced DNA in the host and transfer of the DNA to its progeny
 - III. Identification of DNA with more desirable genes
- (1) I → II → III
 - (2) II → III → I
 - (3) III → II → I
 - (4) III → I → II
- 196.** In PCR, the separation of DNA strands is due to:
- (1) High temperature.
 - (2) Two sets of Primers.
 - (3) *Taq* DNA polymerase.
 - (4) Deoxynucleotides.

- 197.** Which of the following is an **incorrect** match?
- (1) Ringworm infections - *Trichophyton*
 - (2) Pneumonia - *Haemophilus influenzae*
 - (3) Common cold - Retrovirus
 - (4) Amoebic dysentery - *Entamoeba histolytica*
- 198.** A person suffering from a disease caused by *Plasmodium*, experiences recurring chill and fever at the time when:
- (1) the sporozoites released from RBCs are being rapidly killed and broken down inside spleen.
 - (2) the sporozoites released from WBCs are being killed instantaneously.
 - (3) the parasite after its rapid multiplication inside RBCs ruptures them and releases haemozoin.
 - (4) the microgametocytes and megagametocytes are being destroyed by the WBCs.
- 199.** The term 'active immunity' means:
- (1) increasing rate of heart beat.
 - (2) increasing quantity of blood.
 - (3) resistance developed after disease.
 - (4) resistance developed before birth.
- 200.** Which of the following is an **incorrect** match?
- (1) Hepatitis B vaccine - Produced from yeast.
 - (2) Antibody - H₂L₂
 - (3) Filariasis - *Epidermophyton*
 - (4) Malaria - *Plasmodium*

